

Dexter Williams

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Education

University of Wisconsin-Madison

PHD IN INFORMATION SCIENCE

- Advisor: Professor Jodi Schneider.
- Research Interests: Argument Mining, Computational Argumentation, AI and Law.
- 4.0 GPA

Madison, Wisconsin

Aug. 2025 - Present

University of Illinois Urbana-Champaign

PHD IN INFORMATICS

- Advisor: Professor Jodi Schneider.
- 4.0 GPA

Champaign, Illinois

Aug. 2024 - July 2025

University of St Andrews

BSC IN COMPUTER SCIENCE AND PHILOSOPHY (JOINT HONOURS)

- First Class (3.7-3.9 GPA).

St Andrews, United Kingdom

Sep. 2018 - May 2022

Skills

Programming Languages (Strong)	Java, Python, Ada, C, Lua, JavaScript/TypeScript.
Programming Languages (Other)	Prolog, SQL, Haskell, R.
General	Software engineering, Agile, NLP.

Experience

University of Wisconsin-Madison

PROJECT ASSISTANT (PROFESSOR JODI SCHNEIDER)

- Wrote an appendix for a book on argument mining, that documents existing corpora, and developed an website equivalent of the appendix that can be contributed to by the argument mining community.
- Acquired copyright permissions for the 100+ figures used in the book, following up with authors with requests for republication rights.
- Supervised/mentored an undergraduate who provided support for the corpora appendix.

Madison, Wisconsin, United States

Aug. 2025 - Present

Boise State University

RESEARCH INTERN

- Joined the Retico team at Boise State University for 1 month to aid in the research and development of state-of-the-art incremental processing systems.
- Developed two new modules for Retico to handle speech diarization and automatic speech recognition.
- Implemented an end-to-end incremental processing project that performs real-time argument mining on a debate to produce an abstract argumentation framework.

Boise, Idaho, United States

Jun. 2025 - Jul. 2025

University of Illinois Urbana-Champaign

GRADUATE RESEARCH ASSISTANT (PROFESSOR JODI SCHNEIDER)

- Provided feedback and review for a new edition of a book on argumentation mining.
- Read and categorized 100+ papers on argument mining to assist in the writing of a new chapter.
- Explored methods for classifying the multitude of existing argument mining corpora.

Champaign, Illinois, United States

May 2025 - Aug. 2025

University of Illinois Urbana-Champaign

GRADUATE RESEARCH ASSISTANT (PROFESSOR BERTRAM LUDÄSCHER)

- Joined the explainable AI through computational argumentation (XAI-CA) project, which explores how argumentation can aid in explainability.
- Created a program in Alloy to generate minimal abstract argumentation frameworks satisfying certain properties.
- Applied relation-based argument mining to US-legal documents, to automatically construct structured argumentation frameworks.

Champaign, Illinois, United States

May 2025 - Aug. 2025

University of Illinois Urbana-Champaign

TEACHING ASSISTANT

- Assisted in teaching 80+ students introductory formal logic, discrete mathematics, and statistics.
- Ran weekly office hours.
- Graded biweekly assignments.

Champaign, Illinois, United States

Aug. 2024 - May. 2025

Capgemini Engineering

Bath, England, United Kingdom

ASSOCIATE SOFTWARE ENGINEER / CONSULTANT

Jun. 2023 - Jul. 2024

- Joined the development team for a large rail project.
- Assisted in writing and testing the Alloy specification of the project.
- Produced Ada SPARK implementation based on the formal specification.
- Tested the project using a suite of tools developed under the HICLASS Research Programme.
- Provided continuous feedback on the test tools to improve their usability.

Capgemini Engineering

Bath, England, United Kingdom

JUNIOR SOFTWARE ENGINEER / CONSULTANT

Sep. 2022 - Jun. 2023

- Designed a prototype for an engineering design tool.
- Developed the tool, creating an application with a React.js + Redux.js + Plotly.js frontend, and a Python + Flask backend.
- Used Redux Thunk to manage the complicated asynchronous state of the prototype.
- Communicated weekly with the clients of the project, and biweekly with end-users, to gather feedback.
- Organized workload into tickets based on gathered feedback.
- Presented a final demo of the prototype which received positive feedback from the end-user community.

NHS Digital

Leeds, United Kingdom (remote)

DATA SCIENCE AND INNOVATION SUMMER INTERN

Jul. 2021 - Sep. 2021

- Joined the RAP (reproducible analytical pipelines) team at NHS Digital for ten weeks.
- Produced four guides that teach good software engineering practices in order to help teams update their projects to follow RAP guidelines.
- Investigated existing PySpark libraries to find a simple and reproducible way for teams to implement basic metadata input validation in their projects.
- Engaged in daily stand-up meetings, sprint planning, and retrospective meetings. Met and spoke with colleagues outside of work in order to become a good team member.
- Received a recommendation at the end of my internship for meeting/exceeding all my objectives.

University of St Andrews

St Andrews, United Kingdom

UNDERGRADUATE RESEARCH ASSISTANT (PROFESSOR IAN GENT)

Oct. 2020 - Feb. 2021

- Assisted in the progress of a research project which aims to provide human explanations of solutions to Sudoku problems.
- Responsible for investigating variants of Sudoku and testing the effectiveness of the project in solving them.
- Communicated concerns and discussed the project with my supervisor as well as three other members of research staff.

University of St Andrews

St Andrews, United Kingdom

UNDERGRADUATE RESEARCH ASSISTANT (DR. KASIM TERZIC)

Oct. 2019 - May. 2020

- Participated in the development of a program to be used by the Marine Biology department. The program uses machine learning to identify hundreds of seals in large aerial photographs.
- Liaised with my project supervisor every week to seek assistance in understanding and developing the project.
- Updated the output of the program to give it practical use by the given deadline.

Publications

Williams, D. (Forthcoming). That Feels Convincing: A Taxonomy Of Pathotic Argumentation Quality, Proceedings of the 5th European Conference on Argumentation

Forthcoming

Williams, D. (Forthcoming). Computational Assessment of Pathotic Argumentation Quality, Online Handbook of Argumentation for AI, Volume 5

Forthcoming

Williams, D. (Forthcoming). Decorum in the Forum: Emotional Interactions in U.S. Presidential Press Conferences, Arguing Democracy

Forthcoming

Zheng, H., **Williams, D.**, & Ludäscher, B. (2025). Using LLMs to Model Arguments in U.S. Supreme Court Briefs: Preliminary Report, Proceedings of the International Workshop on Translating Natural Legal Language Into Formal Representations

2025